KAIROS

the agentic layer for healthy, original, age-safe content

Technical Design Document

Theme 4 · Media, Content & Digital Experiences

Document	Technical design & architecture specification, v1.0
Platform	Google Cloud Platform — exclusively first-party services
Core pattern	Multi-agent system (Google ADK on Vertex AI Agent Engine) over a shared content-understanding index
Problems addressed	Compulsive consumption (doomscrolling) · repost laundering and copyright erosion · age-inappropriate content reaching children · blunt parental controls
Business model	Creator subscription · studio/agency licence · contextual ad exchange · family plan and platform API
Status	Hackathon submission — MVP scope defined in §15

Contents

1. **Executive summary**
2. **Problem analysis**
   2.1 Doomscrolling is an engineered outcome · 2.2 Repost laundering · 2.3 Age-inappropriate content in child-shaped packaging · 2.4 Parental controls are the wrong shape · 2.5 The commercial problem underneath
3. **Why an agentic system**
   3.1 Autonomy ladder · 3.2 What is deliberately not an agent
4. **Solution overview**
   4.1 Why the two products ship together
5. **The Content Genome**
   5.1 Record schema · 5.2 How each field is produced · 5.3 The originality score
6. **Agent specifications**
   6.1 Orchestrator · 6.2 Shield plane · 6.3 Studio plane
7. **Shield: runtime design**
   7.1 Compiling a sentence into a policy · 7.2 Filtering at retrieval · 7.3 The finite feed · 7.4 The Pace state machine
8. **Studio: runtime design**
   8.1 Demand discovery · 8.2 The structural anti-laundering guarantee · 8.3 Pre-flight clearance · 8.4 End-to-end latency
9. **Data architecture**
   9.1 Store selection · 9.2 Rights ledger schema · 9.3 Public API surface
10. **Security, privacy and compliance**
11. **Safety engineering and evaluation**
12. **Scale and cost**
13. **Business model**
14. **Delivery plan**
15. **Risks**

A. **Appendix A — Google Cloud service inventory**
B. **Appendix B — Glossary**

Figures 1-6 are reproduced at full resolution as separate PNG files alongside this document.

1. Executive summary

Four of the most visible failures in digital media today are usually treated as four separate problems. A teenager loses three hours to a feed. A creator finds their video re-uploaded by an account that added nothing. A nine-year-old is served a Roblox clip whose thumbnail promises a building tutorial and whose content is a simulated shooting. A parent opens a controls panel and is offered an age slider and a daily minute limit, neither of which describes what they actually want.

They are not four problems. They are four symptoms of one missing capability: **nobody on the serving path actually understands the content.** Platforms rank on engagement signals — watch time, replays, comment velocity — which are measurements of the viewer, not descriptions of the video. Every downstream control is therefore built on metadata, keywords and report queues, which is why each one fails in the same way.

Kairos builds the missing layer. Every asset that enters the system is decomposed by a pipeline of Google Cloud vision, speech and reasoning services into a **Content Genome**: a single signed JSON record carrying safety intensities, wellbeing characteristics, educational value, provenance lineage, licence state and a 1,408-dimension multimodal embedding. A mesh of eight specialist agents, supervised by an orchestrator built on the Google Agent Development Kit, reasons over that record to make real decisions on behalf of two distinct users.

- **Kairos Shield** serves families and individuals. It compiles a parent's plain-English content plan into an enforceable policy, filters candidates at retrieval time rather than at render time, and runs a behavioural state machine that ends sessions deliberately instead of letting them run out.
- **Kairos Studio** serves creators and small businesses. It finds under-served demand rather than viral videos to copy, refuses to publish an asset that is only someone else's work, clears rights before publication rather than after a takedown, and routes revenue to originals through an attribution ledger.

The two products are commercially independent but architecturally inseparable: they read and write the same index. Safety is what makes the inventory sellable; provenance is what makes attribution payable; understanding is what makes both possible. That shared index is the defensible asset and the reason this is one platform rather than two apps.

> **THE ONE-SENTENCE VERSION**
>
> Kairos turns every piece of media into a structured, signed, queryable record — and then lets autonomous agents use that record to protect attention, protect children, and pay the people who actually made something.

2. Problem analysis

Each subsection below states the mechanism, not just the complaint. The mechanism determines the design.

2.1 Doomscrolling is an engineered outcome, not a personal failing

Infinite feeds are optimised against a reward signal — session duration or return frequency — using content properties that reliably produce it: fast hooks in the first 1.5 seconds, unresolved narrative tension, mild outrage, and variable-ratio reward scheduling borrowed directly from slot machine design. The user is not choosing badly; the system is solving a different problem from the one the user has.

Existing countermeasures fail because they operate on the wrong variable. A screen-time limit measures **duration**, so it treats forty minutes of a guitar course exactly like forty minutes of rage-bait. A grayscale filter or an app blocker measures **presence**, which the user simply defeats by switching apps. Neither intervention knows anything about what is on the screen.

Two design commitments follow. First, feeds must be **finite by construction** — a session begins with a stated intent and a bounded card set, and it has a last card. Second, intervention must be driven by a **behavioural state machine** reading content properties and interaction shape, not a clock.

2.2 Repost laundering and the erosion of copyright

The dominant low-effort creator strategy is now: ask an algorithm what is trending, download the top result, re-encode it slightly, re-upload it. It works because platform matching is weak against trivial transformations — a 3% crop, a 1.05× speed ramp, a mirror flip, a re-encode at a different bitrate — and because enforcement is retrospective. By the time a takedown lands, the view spike is over and the revenue has been paid.

The instinct is to ban this behaviour. That is the wrong move, and it will not work. The underlying demand is real: people want to participate in a trend without having a studio. The correct design converts laundering into its legitimate cousin — **remix with declared attribution and an automatic revenue split** — and makes that path the easy one while making the laundering path mechanically impossible inside the product.

This requires matching that survives transformation (three independent fingerprint families), matching that happens **before** publication rather than after, and a ledger that can actually pay the original.

2.3 Age-inappropriate content in child-shaped packaging

The specific failure pattern parents describe is not a pornography-filter problem. It is a **packaging** problem. Content is produced inside platforms children are already sanctioned to use, with thumbnails, titles, voices and art styles drawn from children's media, and the harmful material is in the middle of the video rather than the metadata. Community gameplay clips are the clearest case: violent, sexualised or gambling-mechanic content rendered in a blocky, cartoon aesthetic that every keyword filter and thumbnail classifier reads as a children's cartoon.

Related patterns the design must handle explicitly: simulated gambling loops (loot boxes, case openings) presented as gameplay; jump-scare content aimed at provoking a fear response in young children; sexualised avatar behaviour inside otherwise innocuous games; and conversational grooming patterns in accompanying voice chat or comments.

Detection therefore has to be **temporal and multimodal** — sampled across the whole timeline, reading vision, audio, transcript and on-screen text together — and it has to produce **graded intensities per dimension** rather than a single allow/deny bit, because a parent's threshold for cartoon slapstick is not their threshold for simulated firearms.

2.4 Parental controls are the wrong shape

Current controls ask a parent to express their intent in the vocabulary of the platform: an age rating, a minute budget, a category checkbox grid, an app blocklist. What a parent actually holds in their head is a sentence — closer to "football skills and science are fine, no horror, nothing with loot boxes, and tell me if something odd shows up."

The gap between those two vocabularies is where controls fail. Parents under-configure because the interface is tedious, over-block because the categories are coarse, and abandon the tool when it blocks something obviously fine. The design commitment is to accept the sentence and compile it — and then to make every enforcement decision **legible**, so the child is told why in language they can read and the parent can see, diff and roll back every rule.

2.5 The commercial problem underneath

The same absence of understanding suppresses legitimate income. A salaried professional with real expertise cannot easily become a part-time creator, because the unpaid overhead — editing, captioning, reformatting for four platforms, rights clearance, scheduling — exceeds the available hours. Small businesses over-invest in interruptive advertising because the alternative, contextual placement, requires knowing what a video is actually about. And advertisers avoid children's inventory entirely, because they cannot prove it is safe and behavioural targeting of minors is prohibited in most jurisdictions.

A platform that can describe content precisely can address all three: it can automate the creator overhead, sell context instead of surveillance, and certify inventory as provably age-appropriate.

3. Why an agentic system

"Agentic" is used loosely. In this document it means a component that (a) is given a goal rather than a procedure, (b) plans a sequence of actions, (c) invokes tools and observes their results, (d) retains state across the task, and (e) holds a defined level of authority to act without asking. A classifier behind an API endpoint satisfies none of these.

The problem genuinely requires it, for three reasons.

- **The decisions are compositional.** Serving one card requires a policy judgement, a provenance judgement, a wellbeing judgement and a ranking judgement, each with different data, latency and failure characteristics. A single monolithic prompt handling all four is untestable and untunable; separate agents can be evaluated, red-teamed and upgraded independently.
- **The goals legitimately conflict.** The Curator wants engagement, the Guardian wants restriction, the Pace agent wants to end the session, the Growth agent wants reach. These are real tensions that need explicit arbitration, not a blended score. The orchestrator implements a strict precedence rule: **safety outranks wellbeing outranks relevance outranks growth**, always, and every override is logged.
- **The work is multi-step and tool-heavy.** Clearing a video for publication means running four checks, interpreting partial results, deciding whether a licence term covers the intended use, and choosing between three outcomes. That is a plan, not a function call.

3.1 Autonomy ladder

Every agent is assigned an explicit authority level. This is a product decision as much as a safety one: users tolerate autonomy in proportion to reversibility.

Level	Authority	Applies to	Reversal path
L1 — Propose	Produces a recommendation. No state changes until a human approves.	Trend Scout, Production	Not needed; nothing happened
L2 — Act with undo	Acts immediately, but the action is visible and reversible by the affected human.	Guardian, Pace, Rights, Growth	Parent override, user override, creator appeal
L3 — Autonomous	Acts without notification. Confined to actions with no user-visible irreversible effect.	Perception, Curator	Re-run; outputs are derived, not authoritative

> **DESIGN RULE**
> No agent may hold L3 authority over an action that removes money from a creator, restricts a child's access, or publishes to a third-party platform. Those three classes are L1 or L2 by construction, and the constraint is enforced in the orchestrator's action router rather than left to prompt discipline.

3.2 What is deliberately not an agent

Serving is not agentic. When a card is requested, no generative model runs: the Curator executes a filtered approximate-nearest-neighbour query and a rules evaluation against a pre-computed genome. All Gemini inference happens once per asset **at ingest**. This single decision is what makes the latency budget in §8.4 achievable and the cost model in §13 bounded — inference cost scales with the content catalogue, not with traffic.

4. Solution overview

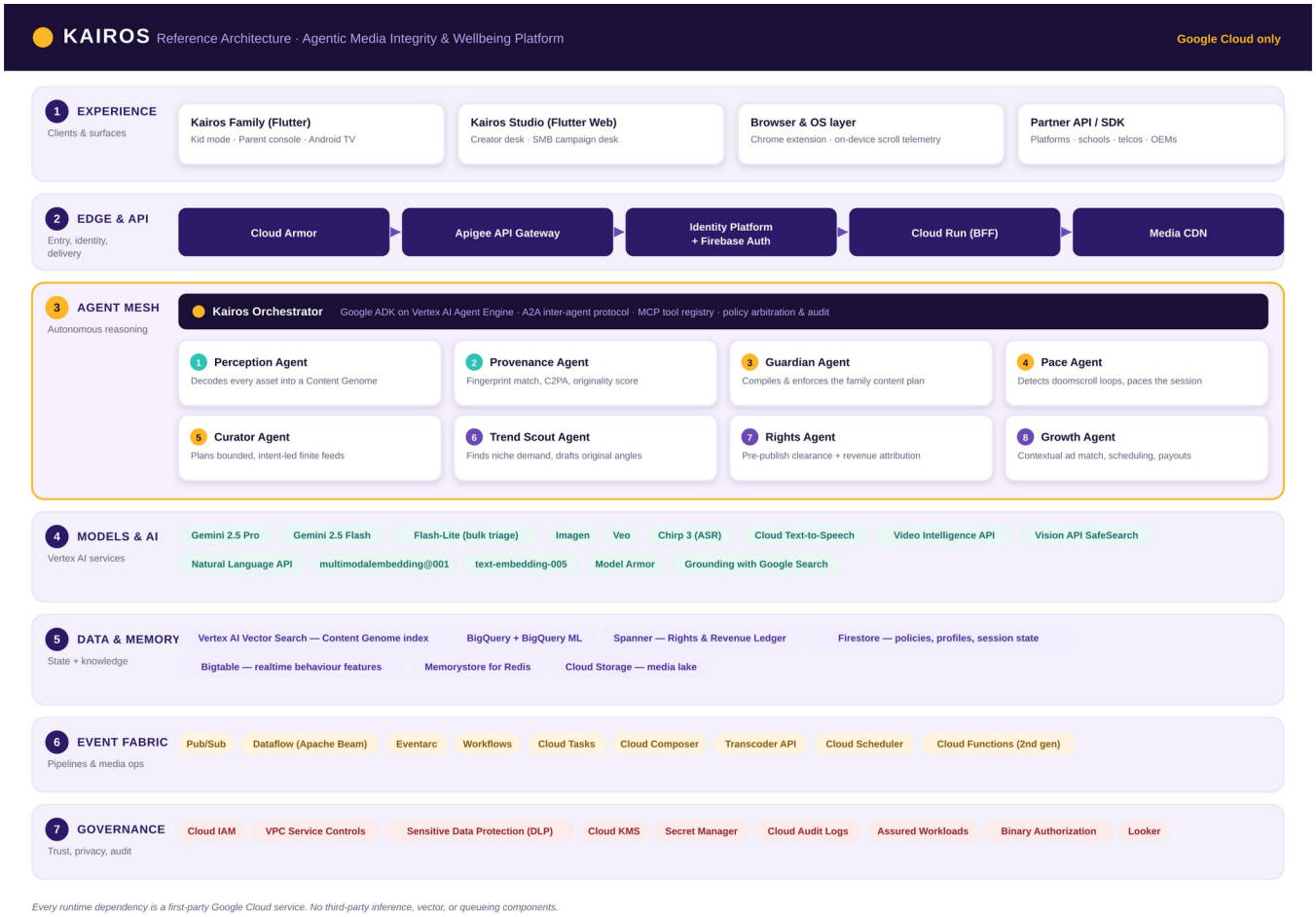


Figure 1 — Kairos reference architecture. Seven layers, all Google Cloud first-party services.

The platform is three things stacked: an understanding layer that produces the Content Genome, an agent mesh that reasons over it, and two products that consume the agents' decisions.

Layer	What it does	Primary Google Cloud services
Experience	Flutter clients for families and creators, a Chrome extension for desktop feeds, Android TV surface, and a partner SDK.	Firebase (Auth, App Check, Remote Config, FCM, Crashlytics), Media CDN
Edge & API	Authentication, rate limiting, partner quotas, WAF, backend-for-frontend aggregation.	Cloud Armor, Apigee, Identity Platform, Cloud Run
Agent mesh	Eight specialist agents under one orchestrator; A2A messaging; tools exposed over MCP.	Vertex AI Agent Engine, Google ADK, Cloud Run (MCP servers)
Models & AI	Multimodal understanding at ingest, generation in Studio, safety filtering everywhere.	Gemini 2.5 Pro / Flash / Flash-Lite, Imagen, Veo, Chirp 3, Cloud TTS, Video Intelligence API, Vision API, Natural Language API, Model Armor
Data & memory	Semantic, episodic, working and analytic memory; the rights ledger.	Vertex AI Vector Search, Firestore, Memorystore, BigQuery, Spanner, Bigtable, Cloud Storage
Event fabric	Ingest fan-out, streaming enrichment, scheduled recomputation, media transcoding.	Pub/Sub, Dataflow, Eventarc, Workflows, Cloud Tasks, Cloud Composer, Transcoder API

Layer	What it does	Primary Google Cloud services
Governance	Least privilege, perimeter control, child-data minimisation, key custody, audit.	Cloud IAM, VPC Service Controls, Sensitive Data Protection, Cloud KMS, Secret Manager, Cloud Audit Logs, Assured Workloads, Looker

4.1 Why the two products ship together

A reviewer will reasonably ask whether a family safety product and a creator tool belong in one system. They do, because of a specific circular dependency:

- Shield needs a supply of content that is known-safe and known-original. Studio produces exactly that, signed at source.
- Studio needs a distribution advantage to be worth paying for. Shield provides it: genuinely original, provably age-appropriate content gets ranked into family feeds and sold into premium brand-safe inventory that competitors cannot certify.
- Both need the Content Genome, and the index gets better with every asset either side processes. The marginal cost of the second product, given the first, is close to the cost of its UI.

5. The Content Genome

The Content Genome is the central data structure. One record per asset, produced once at ingest, versioned, signed, and read by every agent and every partner. If the record is wrong, everything downstream is wrong, so the pipeline that produces it is the most heavily evaluated part of the system.

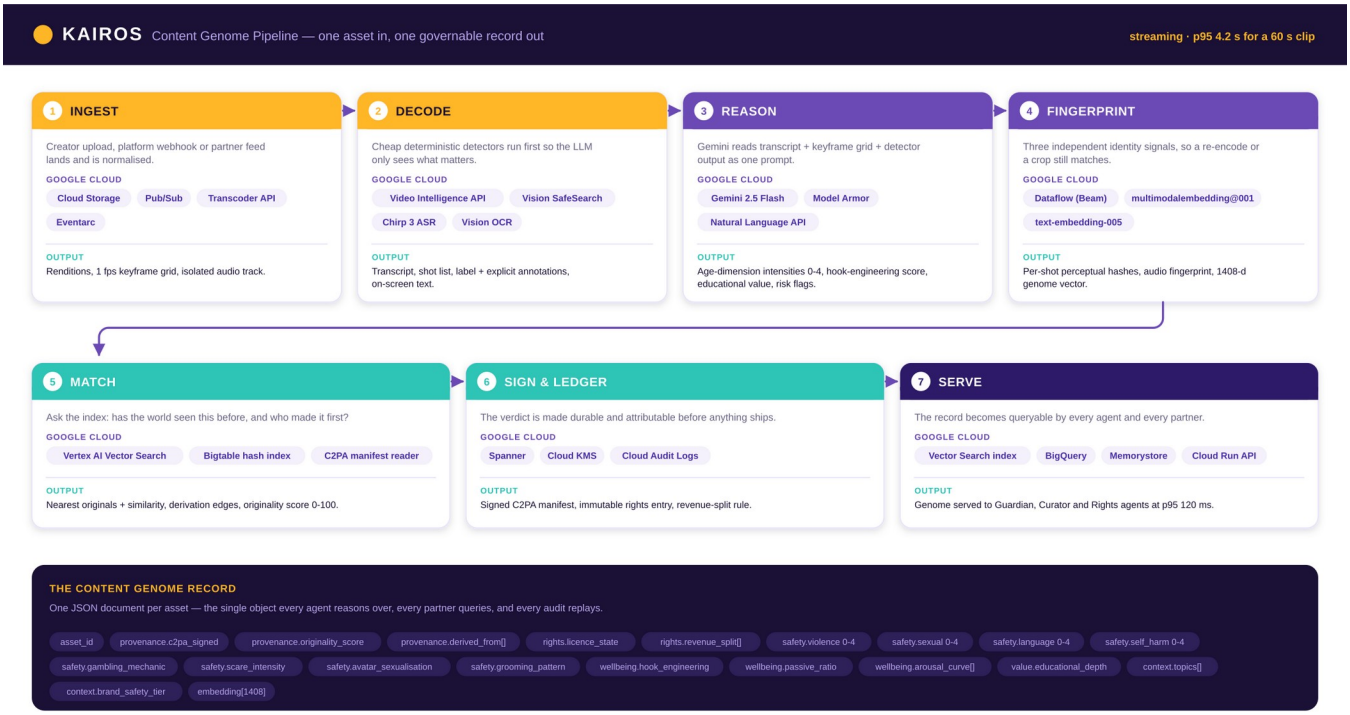


Figure 2 — The Content Genome pipeline. Seven stages, streaming, p95 4.2 s for a 60-second clip.

5.1 Record schema (abridged)

CONTENT GENOME — ABRIDGED RECORD

```
{
  "asset_id": "ast_01J9Q...",            "schema_version": "1.4",
  "source": { "platform": "upload", "creator_id": "crt_7f2", "duration_s": 58 },

  "provenance": {
    "c2pa_signed": true,                 "signer": "kairos-kms-key-3",
    "originality_score": 91,             // 0-100, see 5.3
    "derived_from": [ { "asset_id": "ast_01J2B...", "similarity": 0.38,
                        "relation": "quoted_clip", "declared": true } ]
  },

  "rights": {
    "licence_state": "clear",            // clear | derivative | blocked
    "music": [ { "track_id": "trk_44", "licence": "LIC-2026-8871", "term_ends": "2027-03-01" } ],
    "likeness": { "identifiable_faces": 1, "consent_on_file": true, "minors_present": false },
    "revenue_split": [ { "party": "crt_7f2", "bps": 8500 }, { "party": "crt_1a9", "bps": 1500 } ]
  },

  "safety": {                             // graded 0-4, per timeline segment and aggregate
```

```
    "violence": 1, "sexual": 0, "language": 1, "self_harm": 0,
    "scare_intensity": 0, "gambling_mechanic": false,
    "avatar_sexualisation": false, "grooming_pattern": false,
    "segments": [ { "t0": 31.5, "t1": 38.0, "violence": 3, "note": "simulated firearm" } ]
  },

  "wellbeing": {
    "hook_engineering": 0.71,             // 0-1, density of retention devices
    "passive_ratio": 0.44,                // consumption vs participation affordance
    "arousal_curve": [0.2, 0.5, 0.8, 0.9, 0.4],
    "resolution": true                    // does the narrative close, or dangle
  },

  "value":   { "educational_depth": 3, "skill_transfer": true, "age_fit": [7, 12] },
  "context": { "topics": ["minecraft", "redstone", "tutorial"], "language": "en",
               "brand_safety_tier": "A", "iab_categories": ["IAB9-30"] },
  "embedding": [ /* 1408 floats, multimodalembedding@001 */ ]
}
```

5.2 How each field is produced

Field group	Producer	Method and rationale
safety.*	Video Intelligence API → Vision SafeSearch → Gemini 2.5 Flash	Deterministic detectors run first and are cheap; they localise candidate segments. Gemini then reads only those segments plus the full transcript, which cuts token cost by roughly an order of magnitude versus sending the whole video. Graded output is requested as constrained JSON, not free text.
wellbeing.*	Gemini 2.5 Flash over transcript + shot-change timings	Hook engineering is computed from cut rate in the opening 3 s, presence of an unresolved question, and caption imperative density. Resolution is a direct yes/no judgement on whether the narrative closes.
provenance.*	Dataflow fingerprinting → Vertex AI Vector Search → C2PA reader	Three fingerprint families (per-shot perceptual hash, audio fingerprint, semantic embedding) are matched independently; agreement across families is what produces confidence, because each family fails to a different transformation.
rights.*	Rights Agent + licence registry + Spanner	Licence identifiers are resolved to specific terms, not just presence. The revenue split is written as basis points so it can be executed without floating-point drift.
context.*	Natural Language API + Gemini	Topics and IAB categories drive contextual ad matching without touching any user profile.

5.3 The originality score

A single 0–100 figure, computed from the three fingerprint families plus declaration state. It is deliberately conservative: the cost of wrongly accusing a creator of theft is much higher than the cost of a manual review.

ORIGINALITY SCORING

```
originality = 100
            - 55 * max_shot_hash_overlap        // visual reuse, 0-1
```

```
            - 25 * audio_fingerprint_overlap    // soundtrack / VO reuse, 0-1
            - 20 * (semantic_similarity > 0.93) // near-duplicate framing
            + 30 * declared_and_attributed      // honesty is rewarded, not punished
            + 10 * own_asset_ratio              // share of runtime shot by this creator

verdict = clear       if originality >= 82 and licence_state == "clear"
          derivative  if 45 <= originality < 82 and declared
          blocked     otherwise            // always with a named reason + remix path
```

WHY THE DECLARATION BONUS MATTERS

A creator who says "I am reacting to this clip" scores higher than one who says nothing, even for identical footage. This is the mechanism that converts repost laundering into attributed, paid remix — it makes honesty the cheaper path rather than relying on detection alone.

6. Agent specifications

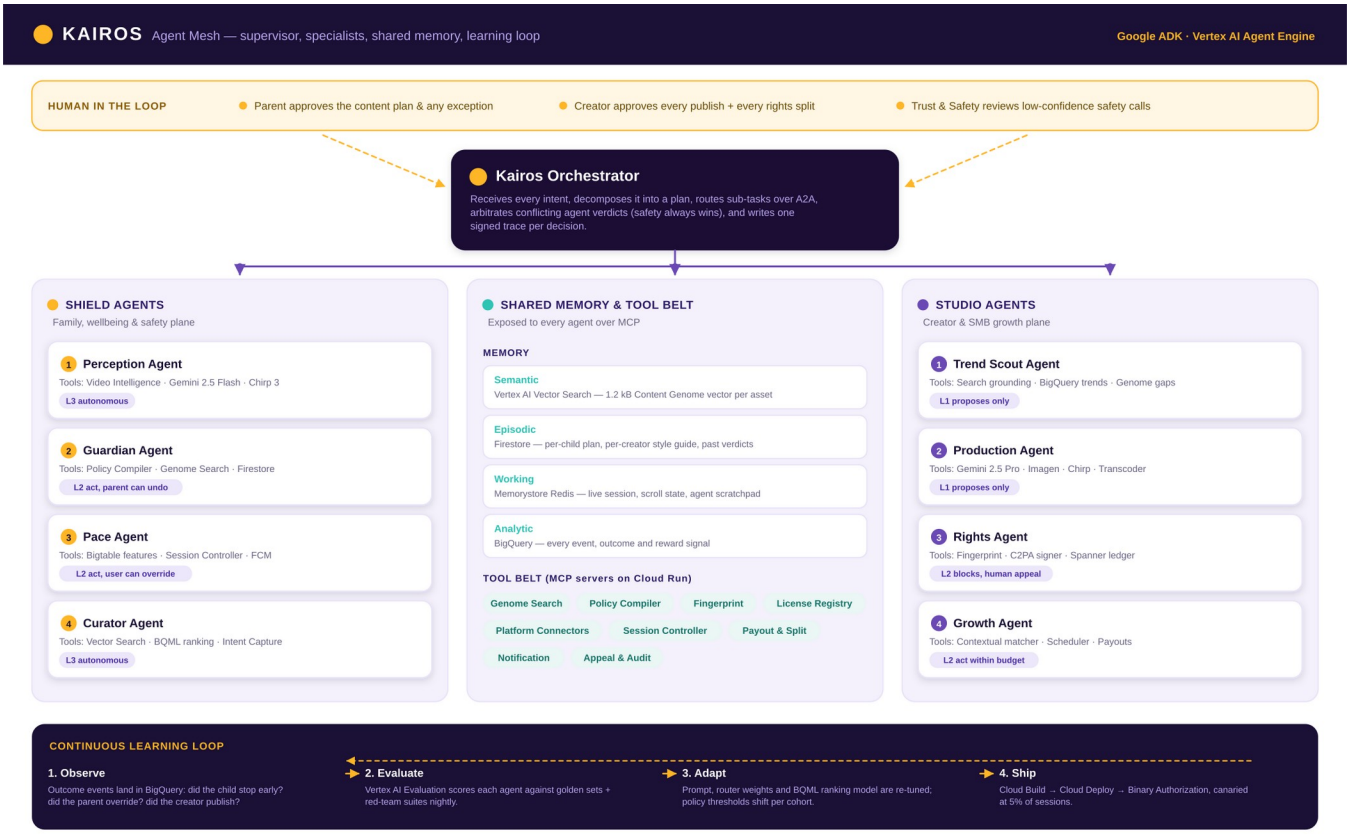


Figure 3 — The agent mesh: supervisor, two specialist planes, shared memory and tool belt, and the nightly learning loop.

All eight agents are implemented with the Google Agent Development Kit and deployed to Vertex AI Agent Engine. They communicate over A2A; every tool is an MCP server on Cloud Run, which means a tool is reusable across agents and independently permissioned in IAM.

6.1 Orchestrator

Property	Specification
Goal	Convert an inbound intent into a plan, route sub-tasks, arbitrate conflicting verdicts, and emit one signed trace per decision.
Arbitration rule	Strict precedence: safety > wellbeing > relevance > growth. A block from Guardian or Rights cannot be outvoted by any other agent at any confidence level.
Escalation	Any safety verdict with confidence below 0.75 is routed to the Trust & Safety queue and the asset is withheld in the interim — fail closed, not open.
Observability	Each decision writes a trace to Cloud Trace and a structured row to BigQuery: inputs, agent verdicts, arbitration result, latency per hop, model versions. Every decision is replayable.

6.2 Shield plane

Agent	Goal & authority	Tools	Primary failure mode and mitigation
Perception	Turn a raw asset into a	Video Intelligence, Vision	Misses harm in an unsampled interval. Mitigated by

Agent	Goal & authority	Tools	Primary failure mode and mitigation
(L3)	complete, valid Content Genome.	SafeSearch, Chirp 3, Vision OCR, Gemini 2.5 Flash	shot-boundary-driven sampling rather than fixed intervals, plus mandatory full-transcript read.
Guardian (L2)	Compile the family content plan into typed rules and evaluate every candidate against it.	Policy Compiler, Genome Search, Firestore	Over-blocks and the family disables it. Mitigated by showing the parent worked examples at compile time, a visible swap log, and a one-tap "this was fine" signal that tunes thresholds.
Pace (L2)	Keep the session aligned to its stated intent; end it deliberately.	Bigtable features, Session Controller, FCM	Feels punitive and gets defeated. Mitigated by silent re-weighting before any visible intervention, and by always offering the user an explicit override that is logged rather than blocked.
Curator (L3)	Plan a bounded, intent-aligned feed with a defined end.	Vector Search, BQML ranking, Intent Capture	Bland feeds; the user leaves for a platform that is more exciting. Mitigated by optimising for completed sessions and next-day return, never for session length.

6.3 Studio plane

Agent	Goal & authority	Tools	Primary failure mode and mitigation
Trend Scout (L1)	Find under-served demand in the creator's niche and draft original angles with evidence.	Grounding with Google Search, BigQuery trend deltas, Genome gap query	Recommends chasing a trend that is already saturated. Mitigated by the gap query — it ranks demand against existing genome coverage, not against view counts.
Production (L1)	Remove the unpaid overhead: script, shot list, captions, dubs, thumbnails, platform cuts.	Gemini 2.5 Pro, Imagen, Chirp, Cloud TTS, Transcoder API	Homogenises voice across creators. Mitigated by a per-creator style profile in episodic memory, and by never auto-publishing — the creator approves every output.
Rights (L2)	Clear or block every asset before publication; register attribution and splits.	Fingerprint, Licence Registry, C2PA signer, Spanner	False accusation of theft. Mitigated by a conservative threshold, three-family agreement, a named appeal path, and human review of every block above a confidence band.
Growth (L2)	Schedule, match contextual advertisers, execute payouts within budget.	Contextual matcher, Scheduler, Payout & Split	Places a brand next to unsuitable context. Mitigated by matching on genome brand-safety tier with advertiser-set floors, and by a kill switch that is an IAM-gated single call.

7. Shield: runtime design

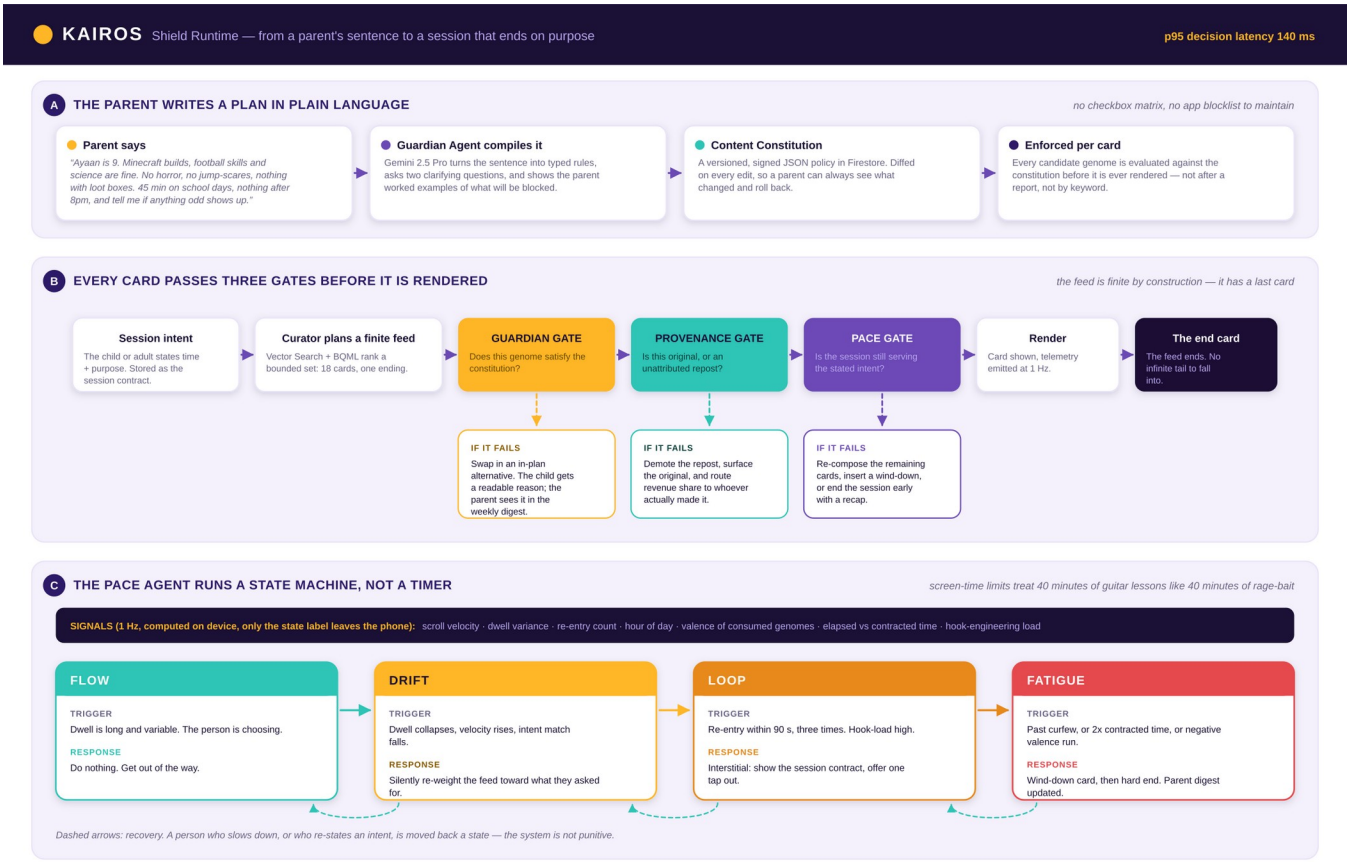


Figure 4 — From a parent's sentence to a session that ends on purpose.

7.1 Compiling a sentence into a policy

The parent writes or speaks a plan. Gemini 2.5 Pro compiles it into a typed policy document, asks at most two clarifying questions, and — critically — shows the parent five real examples of content that would now be blocked and five that would be allowed. Compilation without that preview is how controls end up mis-set.

CONTENT CONSTITUTION — STORED IN FIRESTORE, VERSIONED AND DIFFABLE

```
{ "constitution_id": "con_3k9", "version": 7, "child_id": "chd_88", "age": 9,
  "allow_topics":  ["minecraft.build", "football.skills", "science.explainer"],
  "deny_topics":   ["horror", "gambling_mechanic", "true_crime"],
  "max_intensity": { "violence": 1, "sexual": 0, "language": 1, "scare": 0 },
  "require":       { "min_originality": 60, "resolution": true },
  "time":          { "school_day_minutes": 45, "weekend_minutes": 90, "curfew": "20:00" },
  "notify_parent": ["anomaly", "escalation", "override_used"],
  "signed_by": "par_21", "signed_at": "2026-09-14T19:02:11Z" }
```

7.2 Filtering at retrieval, not at render

This is the most important implementation choice in Shield. Policy predicates are pushed **into** the Vector Search query as filter restrictions, so disallowed content never enters the candidate set. Filtering after retrieval would mean the

system had already ranked, fetched and possibly prefetched material the child must not see — and would leak the existence of that material through timing and cache behaviour.

POLICY-FILTERED RETRIEVAL

```
candidates = vector_search.find_neighbors(
    index        = "content-genome-v4",
    query        = session_intent_embedding,
    neighbor_count = 240,
    filter = [
        Namespace("deny_topic",        deny_tokens,    op="NOT_IN"),
        NumericNamespace("violence",   value=1,        op="LESS_EQUAL"),
        NumericNamespace("sexual",     value=0,        op="LESS_EQUAL"),
        NumericNamespace("originality", value=60,      op="GREATER_EQUAL"),
        Namespace("gambling_mechanic", ["false"],      op="IN"),
    ])
# Guardian then re-evaluates the survivors against the full constitution,
# because filters express thresholds but not conditional or temporal rules.
```

7.3 The finite feed

The Curator returns a fixed card set — typically 18 for a 20-minute contract — ordered so that the highest-value item is not first (which would train hook-seeking) and with an **end card** pinned last. The end card is a real UI state, not a spinner: it shows what the session contained, what was learned or skipped, and offers exactly two actions — finish, or extend by a named amount with a new contract. Extension is always allowed; it is simply never the default and never silent.

7.4 The Pace state machine

Four states, driven by signals computed on-device at 1 Hz. Only the derived state label leaves the device — raw scroll traces, dwell times and interaction data never reach Google Cloud, which matters both for privacy posture and for regulatory exposure on minors' data.

State	Entry condition	System response
FLOW	Dwell long and variable; intent match high.	No action. Log only.
DRIFT	Dwell variance collapses, scroll velocity rises, intent match falls below 0.5.	Silently re-weight remaining cards toward the stated intent. No UI change.
LOOP	Three re-entries within 90 s, or hook-engineering load above 0.7 sustained for 5 cards.	Interstitial: restate the session contract, offer a one-tap exit, pause autoplay.
FATIGUE	Past curfew, or 2× contracted time, or a sustained negative-valence run.	Wind-down card, then hard end. Parent digest updated. Override available and logged.

Transitions in the recovery direction are always permitted: a user who slows down or restates an intent moves back a state. The system is designed to be non-punitive, because a punitive wellbeing feature is uninstalled within a week.

8. Studio: runtime design

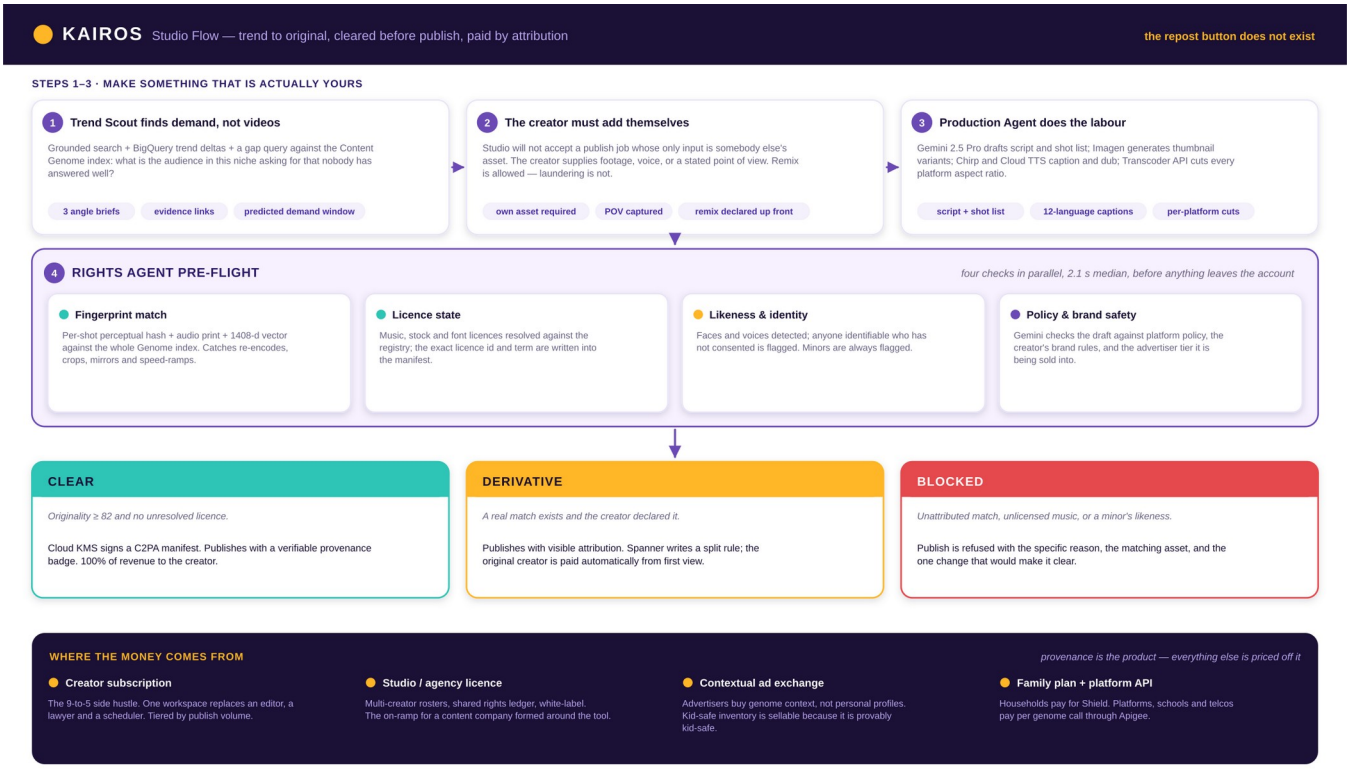


Figure 5 — Trend to original, cleared before publish, paid by attribution.

8.1 Demand discovery instead of video copying

Trend Scout does not return videos to imitate. It runs three queries and intersects them: a grounded search for rising questions in the niche, a BigQuery delta over topic velocity in the genome index, and a **coverage gap query** that finds topics with rising demand and thin high-quality supply. The output is three angle briefs, each with evidence links and a predicted demand window — an input to a creator's judgement, not a replacement for it.

8.2 The structural anti-laundering guarantee

Studio has no repost path. A publish job is rejected at the API contract level unless it carries at least one creator-originated asset or a declared and attributed derivation. This is enforced in the Rights Agent's action router, not in a prompt, so it cannot be talked around. The product consequence is that the fastest way to participate in a trend inside Kairos is remix-with-attribution — which pays the original creator automatically from the first view.

8.3 Pre-flight clearance

Four checks run in parallel, median 2.1 s: fingerprint match against the whole genome index, licence resolution for music, stock and fonts, likeness and identity detection, and a policy and brand-safety read. The verdict is one of three, and a block always names the specific reason, the matching asset, and the single change that would clear it.

8.4 End-to-end latency

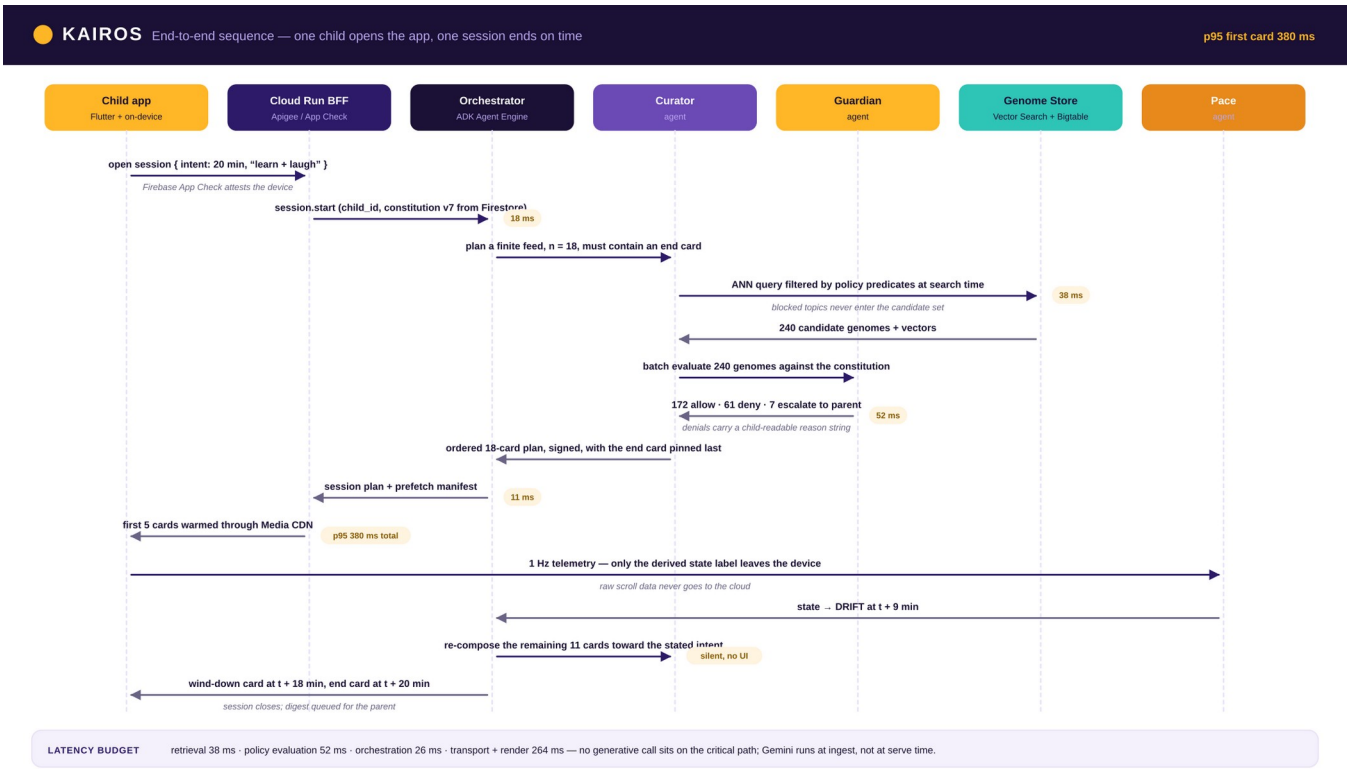


Figure 6 — Sequence for a Shield session, with the latency budget.

The budget holds because no generative call is on the serving path. Retrieval is 38 ms, policy evaluation 52 ms, orchestration 26 ms, and the remainder is transport and render. Gemini inference is amortised across the catalogue at ingest.

9. Data architecture

9.1 Store selection

Store	Holds	Why this store and not another
Cloud Storage	Source media, renditions, keyframes, transcripts.	Object lifecycle rules move keyframes to Nearline after 30 days; originals are the only thing that must be durable forever.
Vertex AI Vector Search	Genome embeddings with filterable namespaces.	The only store here that does filtered ANN at the scale and latency the retrieval-time filtering design in §7.2 requires.
Firestore	Content constitutions, creator style profiles, session state, agent episodic memory.	Document shape matches the data, real-time listeners drive the client, and per-document security rules map cleanly onto per-child access control.
Spanner	Rights ledger: derivation edges, licence terms, revenue splits, payouts.	This is the one place that needs external consistency and strong transactions across regions — it is money and legal attribution. Eventual consistency here would be a defect.
Bigtable	Per-session behavioural features, fingerprint hash index.	Single-digit-millisecond reads on a wide, sparse, append-heavy key space; the hash index is a classic row-key lookup.
BigQuery	Every event, every decision trace, training and evaluation sets.	Analytical warehouse plus BQML for the ranking model, so ranking trains where the data already lives.
Memorystore	Working memory: live session, agent scratchpad, rate limits.	Sub-millisecond, ephemeral by design — nothing here should survive a restart.

9.2 Rights ledger schema (Spanner, abridged)

SPANNER DDL

```
CREATE TABLE Asset (
  asset_id        STRING(36) NOT NULL,
  creator_id      STRING(36) NOT NULL,
  originality     INT64 NOT NULL,
  c2pa_manifest   BYTES(MAX),
  published_at    TIMESTAMP OPTIONS (allow_commit_timestamp=true),
) PRIMARY KEY (asset_id);

CREATE TABLE Derivation (
  asset_id        STRING(36) NOT NULL,    -- the new work
  source_id       STRING(36) NOT NULL,    -- the work it draws on
  similarity      FLOAT64 NOT NULL,
  relation        STRING(24) NOT NULL,    -- quoted_clip | remix | cover | repost
  declared        BOOL NOT NULL,
) PRIMARY KEY (asset_id, source_id),
  INTERLEAVE IN PARENT Asset ON DELETE CASCADE;

CREATE TABLE Split (
  asset_id        STRING(36) NOT NULL,
  party_id        STRING(36) NOT NULL,
  bps             INT64 NOT NULL,         -- basis points, must sum to 10000
```

```
  agreed_at       TIMESTAMP NOT NULL,
) PRIMARY KEY (asset_id, party_id),
  INTERLEAVE IN PARENT Asset ON DELETE CASCADE;
```

9.3 Public API surface (via Apigee)

REST SURFACE

```
POST /v1/assets                 → submit for genome extraction  (202 + job id)
GET  /v1/assets/{id}/genome     → full genome record            (scoped by IAM)
POST /v1/assets/{id}/clearance  → run pre-flight, return verdict
POST /v1/policies/compile       → { "plan": "<plain text>" } → constitution + examples
POST /v1/sessions               → { intent, minutes } → finite card plan
POST /v1/sessions/{id}/telemetry→ { state_label, t } (no raw behavioural data accepted)
GET  /v1/rights/{id}/lineage    → derivation graph + current splits
POST /v1/inventory/match        → advertiser context match against genome tiers
```

10. Security, privacy and compliance

A product that processes children's media and behavioural data has a compliance surface that is a design input, not a legal review at the end.

Control	Implementation
Data minimisation for minors	Behavioural signals are derived on-device; only a four-value state label is transmitted. No raw scroll, dwell or interaction stream is stored in the cloud for any user under 18.
No behavioural advertising to minors	Ad matching runs against the genome vector of the content, never against a user profile. This is architectural — the ad-matching service has no IAM grant to any user-profile store, so the prohibited join is impossible rather than merely forbidden.
PII handling	Sensitive Data Protection inspects and de-identifies transcripts and comments before they enter BigQuery. Face and voice vectors are stored only where consent is recorded.
Perimeter	VPC Service Controls around all data stores and Vertex AI; egress restricted to the CDN and Apigee. Private Service Connect for all internal service-to-service calls.
Key custody	Cloud KMS with CMEK on Cloud Storage, BigQuery, Spanner and Firestore. C2PA signing keys held in Cloud HSM and never exported.
Access	Cloud IAM with per-agent service accounts; an agent holds only the grants its tool belt requires. Deployment gated by Binary Authorization.
Prompt-injection defence	Model Armor on every prompt path. Content retrieved from third-party platforms is treated as untrusted data and never as instructions — agents run with a fixed system contract and a tool allowlist.
Audit	Cloud Audit Logs plus an immutable decision log in BigQuery. Any enforcement decision can be replayed with the exact model versions and inputs that produced it.
Residency and regime	Assured Workloads for regional data boundaries; the design targets COPPA, GDPR / GDPR-K, the UK Age Appropriate Design Code, and the EU DSA minor-protection provisions.

REGULATORY NOTE

The specific obligations vary considerably by jurisdiction and are changing quickly, particularly around age assurance and minors' data. The controls above are the engineering posture; a jurisdiction-by-jurisdiction legal review is required before launch in any market and is out of scope for this document.

11. Safety engineering and evaluation

An AI system that decides what a child may watch and whether a creator has stolen something needs an evaluation regime proportional to those stakes.

11.1 Evaluation sets

- **Golden safety set** — 5,000 hand-adjudicated clips spanning every safety dimension, deliberately over-weighted toward the hard cases from §2.3: child-styled packaging over adult content, gambling mechanics, jump-scares, avatar sexualisation.
- **Adversarial provenance set** — every asset present in nine transformations: crop, mirror, 0.9×–1.1× speed, re-encode, letterbox, colour shift, overlay, partial clip, and audio replacement. Target: recall above 0.95 at a false-positive rate under 0.01.
- **Policy compilation set** — 800 real parent sentences with expert-labelled rule expansions, measuring whether the compiled constitution matches stated intent.
- **Red-team suite** — prompt injection embedded in transcripts, on-screen text and comments; jailbreak attempts against the Guardian; attempts to get the Rights Agent to clear known-infringing material.

11.2 Metrics that matter

Metric	Definition	Target
Harmful-serve rate	Fraction of rendered cards later adjudicated as policy violations.	< 0.05% at p95
False-block rate	In-policy content incorrectly blocked, on the golden set.	< 2%
Parent override rate	Overrides per 100 enforcement actions — a proxy for compilation quality.	< 5 and falling
Session completion	Sessions ending at the end card rather than being abandoned.	> 60%
Loop entry rate	Sessions reaching the LOOP state at least once.	Falling month over month
Provenance recall	Transformed duplicates correctly matched (adversarial set).	> 0.95
Attribution payout	Share of derivative revenue actually reaching original creators.	> 98% within 30 days
Brand-safety incidents	Advertiser-reported placements below their stated tier floor.	Zero tolerance; each triggers review

THE METRIC WE REFUSE TO OPTIMISE

Session length and daily time-in-app are tracked as diagnostics and are never targets. An honest wellbeing product must be willing to reduce its own engagement numbers, and the incentive to quietly stop doing so is the single largest long-term risk to this product's integrity.

12. Scale and cost

12.1 Scaling characteristics

Dimension	Behaviour	Mechanism
Catalogue growth	Linear in assets, fully parallel.	Pub/Sub fan-out into Dataflow; per-asset processing is independent, so throughput scales with worker count.
Traffic growth	Sub-linear in cost — no inference on the serving path.	Cloud Run autoscaling; hot genomes cached in Memorystore; Media CDN absorbs delivery.
Index growth	Logarithmic query cost.	Vertex AI Vector Search sharded by locale and age band, so a family query searches a fraction of the corpus.
Agent concurrency	Independent per plane.	Each agent is a separately scaled Agent Engine deployment; Shield load and Studio load do not contend.

12.2 Unit cost model

Costs are dominated by one-time ingest, which is the desirable shape: the marginal cost of an additional viewer is small. The model below is a structure to be filled with current list prices at build time — **the coefficients must be verified against live Google Cloud pricing, which changes and which this document does not attempt to quote.**

UNIT ECONOMICS STRUCTURE

```
cost_per_asset  =  transcode(duration)
                +  video_intelligence(duration)
                +  chirp_asr(duration)
                +  gemini_flash(transcript_tokens + keyframe_tokens)
                +  embedding(1 call)
                +  dataflow(fingerprint CPU-seconds)
                +  storage(renditions, monthly)

cost_per_session = vector_search(1 filtered ANN query)
                 + firestore(≈6 reads)
                 + bigtable(≈40 writes)
                 + cloud_run(≈120 ms CPU)
                 + cdn_egress(bytes served)

Levers, in order of impact:
  1. Flash-Lite triage before Flash        — skip full reasoning on ~70% of assets
  2. Shot-boundary sampling not fixed-rate — fewer keyframe tokens, better recall
  3. Genome caching in Memorystore         — repeat views cost almost nothing
  4. Committed-use discounts on Dataflow and Vector Search
```

13. Business model

Stream	Buyer	What they are actually paying for
Creator subscription	The salaried professional building a side income	Elimination of the unpaid overhead — editing, captioning, reformatting, clearance, scheduling — that is the real reason most people never start. Tiered by publish volume.
Studio / agency licence	A small content company, or a creator collective becoming one	Multi-creator rosters on a shared rights ledger, white-labelling, and consolidated payouts. This is the path from "person with a phone" to "registered content business", and the ledger is what makes the business auditable enough to raise money against.
Contextual ad exchange	SMBs and brands	Placement bought against content context rather than user surveillance. Two things are being sold that competitors cannot easily offer: provably brand-safe inventory, and legally clean access to family audiences.
Family plan	Households	Parental control that works on content rather than clocks, plus a weekly digest that is genuinely informative.
Platform / institution API	Platforms, schools, telcos, OEMs	Per-call genome access through Apigee. The margin here is high because the index is already built for the other four streams.

13.1 Why this is defensible

- The Content Genome index compounds. Every asset processed for any customer improves provenance matching for every other customer — a direct network effect on an asset that competitors must rebuild from zero.
- Provably safe inventory is a regulatory arbitrage that is legal, durable and getting more valuable as minor-protection rules tighten. Behavioural targeting of children is closing off; contextual targeting is the only remaining compliant route to that audience.
- The attribution ledger creates switching costs for exactly the creators worth keeping: anyone receiving derivative revenue through Kairos loses that income by leaving.

14. Delivery plan

14.1 Hackathon MVP — what actually gets built

The full system is a multi-quarter build. The demonstrable vertical slice is deliberately narrow and covers all four stated problems end to end on a seeded catalogue of roughly 2,000 clips.

In scope	Deferred
Perception, Guardian, Pace, Curator and Rights agents (5 of 8) on ADK + Agent Engine	Trend Scout, Production and Growth agents
Genome pipeline for short-form video, English only	Live streams, long-form, multilingual dubbing
Policy compilation from a parent sentence, with the worked-example preview	Voice-driven policy editing, multi-child households
Filtered retrieval, finite feed, end card, full Pace state machine	On-device model; MVP computes state in Cloud Run with a documented privacy gap
Fingerprint matching across three families + originality score + three-way verdict	C2PA signing and the Spanner payout execution path (schema built, payouts stubbed)
Parent digest in Looker Studio	Advertiser exchange UI

14.2 Beyond the hackathon

Phase	Focus	Exit criterion
0–3 months	Harden the genome pipeline; 5,000-clip golden set; first 100 families in closed beta.	Harmful-serve rate below target on the golden set; parent override rate under 10.
3–6 months	Studio launch with Trend Scout and Production; C2PA signing live; payouts executing.	First attributed revenue split paid to an original creator.
6–12 months	Contextual exchange with launch advertisers; platform API through Apigee.	Two platform integrations live; brand-safety incident count at zero.
12+ months	On-device Gemini Nano for behavioural state; multilingual; institutional deployments.	No raw behavioural data leaves any device, anywhere.

15. Risks

Risk	Severity	Mitigation
Model misclassifies harmful content as safe	Critical	Multi-detector agreement, fail-closed on low confidence, human review queue, and a published incident process. Accepted as never fully eliminable — the target is a measured, falling rate, not zero.
False accusation of plagiarism damages a creator	High	Conservative threshold, three-family agreement requirement, human review above a confidence band, and a named appeal path with a service-level response time.
Parents over-block and children route around the product	High	Worked-example preview at compile time, visible swap log, one-tap "this was fine" feedback, and age-graded autonomy that widens as the child gets older.
Wellbeing features suppress engagement metrics and get quietly weakened	High	Session length is never a target and is excluded from all bonus and OKR structures; the end card is a contractual product guarantee, not a setting.
Platforms restrict API access	Medium	The genome pipeline works on any asset the user can supply. Kairos is positioned as complementary infrastructure and offered to platforms as a paid API rather than as a scraper.
GCP concentration risk	Medium	Accepted deliberately for delivery speed and integration depth. Portability is preserved where it is cheap: containers on Cloud Run, Beam for pipelines, standard SQL, open C2PA manifests.
Cost per asset exceeds the model at scale	Medium	Flash-Lite triage tier, shot-boundary sampling, aggressive genome caching, and a hard per-tenant ingest budget enforced in Apigee.

A. Appendix A — Google Cloud service inventory

Service	Used for
Vertex AI — Gemini 2.5 Pro	Policy compilation, script and shot-list generation, brand-safety reasoning
Vertex AI — Gemini 2.5 Flash	Per-asset safety and wellbeing reasoning at ingest
Vertex AI — Gemini 2.5 Flash-Lite	Bulk triage tier: decides which assets need full reasoning
Vertex AI — Imagen	Thumbnail variant generation in Studio
Vertex AI — Veo	Optional b-roll generation for Production Agent
Vertex AI — Chirp 3	Speech recognition for transcripts and dubbing
Cloud Text-to-Speech	Voice-over and accessibility narration
Vertex AI — multimodalembedding@001	Genome embedding vector (1,408 dims)
Vertex AI — text-embedding-005	Transcript and intent embeddings
Vertex AI Vector Search	Filtered ANN retrieval over the Content Genome index
Vertex AI Agent Engine + Google ADK	Hosting and orchestration of the eight agents
Vertex AI Evaluation	Nightly agent scoring against golden and adversarial sets
Model Armor	Prompt-injection and jailbreak defence on every prompt path
Video Intelligence API	Shot change, label, explicit content and object tracking
Cloud Vision API	SafeSearch and OCR on keyframes
Cloud Natural Language API	Entity and sentiment analysis on transcripts and comments
Transcoder API	Rendition generation and per-platform aspect-ratio cuts
Cloud Storage	Media lake: sources, renditions, keyframes
BigQuery + BigQuery ML	Event warehouse, decision log, ranking model training
Spanner	Rights ledger: derivations, licences, splits, payouts
Firestore	Constitutions, profiles, session state, episodic agent memory
Bigtable	Behavioural feature store and fingerprint hash index
Memorystore for Redis	Working memory, hot genome cache, rate limiting
Pub/Sub	Ingest fan-out and the event backbone
Dataflow	Streaming enrichment and fingerprint computation
Cloud Composer / Workflows / Eventarc / Cloud Tasks	Batch orchestration, event routing, deferred work
Cloud Run	Stateless services, BFF, MCP tool servers
GKE Autopilot	Long-running fingerprint workers
Apigee	Partner API management, quotas, monetisation
Cloud Armor	WAF and DDoS protection
Identity Platform / Firebase Auth	Household and creator identity, App Check attestation
Media CDN	Edge delivery of prefetched cards

Service	Used for
Cloud IAM / VPC Service Controls / Binary Authorization	Least privilege, perimeter, supply-chain control
Sensitive Data Protection (DLP)	PII inspection and de-identification
Cloud KMS / Cloud HSM / Secret Manager	CMEK, C2PA signing keys, credential custody
Cloud Logging / Monitoring / Trace / Error Reporting	Observability and decision tracing
Looker / Looker Studio	Parent digest, advertiser dashboard, internal analytics
Cloud Build / Artifact Registry / Cloud Deploy	CI/CD with canary rollout
Assured Workloads	Regional data boundaries and compliance regime controls

B. Appendix B — Glossary

Term	Meaning
Content Genome	The signed JSON record produced for every asset: safety, wellbeing, provenance, rights, context and embedding.
Content Constitution	A family's content plan compiled into typed, versioned, signed rules stored in Firestore.
End card	The terminal UI state of a finite feed. A real screen, not a loading state.
Session contract	The stated intent and duration a user agrees at session start; the Pace agent measures against it.
Originality score	A 0–100 provenance figure combining three fingerprint families with declaration state.
Derivation edge	A recorded relationship in the Spanner ledger between a new work and a work it draws on.
Hook engineering	A 0–1 measure of retention-device density: cut rate, unresolved questions, imperative captions.
A2A	Agent-to-Agent protocol used for inter-agent messaging within the mesh.
MCP	Model Context Protocol; every Kairos tool is an MCP server, independently permissioned in IAM.
C2PA	Open content-provenance standard; Kairos signs manifests with keys held in Cloud HSM.